ALGEBRA 1 · UNIT 2: SOLVING LINEAR EQUATIONS

Escape Room Designer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students break out of "The Vault" by solving a chain of equation locks where each answer feeds the next, investigate trick locks with no solution or infinitely many solutions, and then design and test their own escape room.

Because each lock depends on the previous answer, students get immediate feedback: a wrong answer makes the next lock 'not work' (it gives an unreasonable result). This builds the habit of checking solutions. Part 2 extends to word problems and special cases (no solution, identities). Part 3 flips the role so students must construct equations with chosen solutions.

Standards Addressed

- A-REI.1 Explain each step in solving an equation
- A-REI.3 Solve linear equations in one variable, including coefficients represented by letters
- A-CED.1 Create equations and use them to solve problems
- A-CED.4 Rearrange formulas (Honors)

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: set a timer and let groups race through Part 1. Discuss strategies for checking answers.
Day 2	Part 2 word locks and trick locks. Whole-class discussion: What does 'no solution' look like? What about 'all real numbers'?
Day 3	Part 3: groups build their own rooms, then swap with another group to test them.

Materials

- Student handout
- Lock template (included in Part 3)
- Envelopes or a digital form to hold lock answers
- Optional: real combination locks or a free online lock builder

How the Three Versions Differ

Level	What changes
CP2	Four-lock chain with one- and two-step equations, distributive property, and variables on both sides; trick locks identified with support.
CP1	Locks include fractions, variables on both sides, and a decoy set with no-solution and identity equations; word locks include consecutive integers and age problems.
Honors	Locks include rational equations, a literal equation, and parameter locks (choose k so the equation has infinitely many or no solutions); students prove when ax + b = cx + d has one solution.

Watch For (Common Misconceptions)

- Distributing to only the first term inside the parentheses.
- Thinking 0 = 0 means the answer is 0, rather than all real numbers.
- Forgetting to check that a designed lock really has the intended answer.

Extension Ideas

- Turn the best student room into a real class event with lock boxes.
- Require one lock to use an inequality (previews Unit 3).
- Time each group and graph completion times.

Answer Key – CP2

Part 1: Break Out of The Vault

#	Task	Answer
1	Lock 1: Solve 3x + 7 = 31.	3x = 24 → x = 8
2	Lock 2: Let A be your Lock 1 answer. Solve 2(y − A) = 10.	2(y − 8) = 10 → y − 8 = 5 → y = 13
3	Lock 3: Let B be your Lock 2 answer. Solve 5z − B = 2z + 2.	5z − 13 = 2z + 2 → 3z = 15 → z = 5
4	Lock 4: Use your Lock 3 answer for z. Solve (w + z) ÷ 3 = 4.	(w + 5) ÷ 3 = 4 → w + 5 = 12 → w = 7
5	Write the final four-number code. Check one of your answers by substituting it back into its equation.	Code 8 – 13 – 5 – 7. Check: 3(8) + 7 = 31 ✓

Part 2: Word Locks & Trick Locks

#	Task	Answer
6	The keypad lock says: "Triple my number, then subtract 4. The result is 20." What is the number?	3n − 4 = 20 → n = 8
7	The clock lock: "Twice the hour plus 3 equals the hour plus 10." What hour opens the lock?	2h + 3 = h + 10 → h = 7 (7 o'clock)
8	A teammate solved 4(x − 2) = 20 like this: 4x − 2 = 20, 4x = 22, x = 5.5. Find the mistake and solve correctly.	The 4 was not distributed to the 2: 4x − 8 = 20 → 4x = 28 → x = 7
9	Two trick locks: (a) 2(x + 3) = 2x + 6 and (b) 2x + 5 = 2x + 1. Simplify each. Which lock opens with ANY number? Which lock can NEVER open?	(a) 2x + 6 = 2x + 6 → 6 = 6, any number works. (b) 5 = 1 is false, so it can never open (no solution).

Part 3: Design Your Own Room

#	Task	Answer
10	Pick a theme and write a 2–3 sentence story for your escape room.	Answers vary.
11	Choose your three codes first. Then build: Lock 1 = a two-step equation; Lock 2 = an equation with parentheses that uses the Lock 1 code; Lock 3 =…	Answers vary; check each lock with the group's codes.
12	Write your answer key and show that each code checks. Then trade with another group and record how long it took them.	Answers vary.

Answer Key – CP1

Part 1: Break Out of The Vault

#	Task	Answer
1	Lock 1: Solve 4(x − 3) + 2 = 2x + 8.	4x − 12 + 2 = 2x + 8 → 4x − 10 = 2x + 8 → 2x = 18 → x = 9
2	Lock 2: Let A be the Lock 1 code. Solve ½(y + A) = y − 6.	½(y + 9) = y − 6 → ½y + 4.5 = y − 6 → 10.5 = ½y → y = 21
3	Lock 3: Let B be the Lock 2 code. Solve 3(z − 2) − (z + 4) = B − 7.	3z − 6 − z − 4 = 21 − 7 → 2z − 10 = 14 → 2z = 24 → z = 12
4	Lock 4 is a decoy. Only one of these locks has a single code: (a) 5w − 2(w + 4) = 3w + 1, (b) 6w + 4 = 2(3w + 2), (c) 7w − 12 = 3w + 16. Solve all…	(a) 3w − 8 = 3w + 1 → −8 = 1, no solution. (b) 6w + 4 = 6w + 4, infinitely many. (c) 4w = 28 → w = 7. Code 7.
5	Write the final code.	9 – 21 – 12 – 7

Part 2: Word Locks

#	Task	Answer
6	The sum of three consecutive integers is 72. The code is the smallest integer.	n + (n + 1) + (n + 2) = 72 → 3n + 3 = 72 → n = 23 (23, 24, 25). Code 23.
7	The game master is 3 times as old as her nephew. In 12 years, she will be twice as old as he will be. How old is each person now?	3n + 12 = 2(n + 12) → 3n + 12 = 2n + 24 → n = 12. Nephew 12, game master 36.
8	A rectangular trapdoor has a length 4 inches more than twice its width. Its perimeter is 50 inches. Find its dimensions.	2(2w + 4) + 2w = 50 → 6w + 8 = 50 → w = 7 in, length = 18 in
9	A teammate solved 6 − 2(x + 5) = 3x + 1 and got x = 1. Without solving, show how substitution proves the answer is wrong. Then solve correctly.	Substitute x = 1: left 6 − 2(6) = −6; right 3(1) + 1 = 4; −6 ≠ 4. Correct: 6 − 2x − 10 = 3x + 1 → −2x − 4 = 3x + 1 → −5 = 5x → x = −1

Part 3: Design Your Own Room

#	Task	Answer
10	Write a theme and story. Choose your codes first.	Answers vary.
11	Build four chained locks: one with fractions, one with variables on both sides, one word problem, and one decoy set (one no-solution lock, one…	Answers vary; check each lock and the decoy set.
12	Write a full answer key. Trade rooms with another group, then record their time and any lock that caused trouble. Revise one lock.	Answers vary.

Answer Key – Honors

Part 1: Break Out of The Vault

#	Task	Answer
1	Lock 1: Solve (2x + 1) ÷ 3 = (x + 4) ÷ 2.	Multiply both sides by 6: 2(2x + 1) = 3(x + 4) → 4x + 2 = 3x + 12 → x = 10
2	Lock 2: The lock uses the formula d = rt + k. Solve the formula for t. Then find t when d = 94, r = 7, and k is your Lock 1 code.	t = (d − k) ÷ r; t = (94 − 10) ÷ 7 = 84 ÷ 7 = 12
3	Lock 3 (parameter lock): The code is the value of k that makes 3(x − k) = 3x − 12 true for every x. Find k and explain.	3x − 3k = 3x − 12 for every x requires −3k = −12, so k = 4.
4	Lock 4: Solve 0.4(5w − 20) = 1.5w − 2 + 0.25w. Then write the four-number code.	2w − 8 = 1.75w − 2 → 0.25w = 6 → w = 24. Code 10 – 12 – 4 – 24

Part 2: Advanced Locks

#	Task	Answer
5	The coin vault holds 30 coins, all dimes and quarters, worth $5.10. The code is the number of quarters.	0.10(30 − q) + 0.25q = 5.10 → 3 + 0.15q = 5.10 → 0.15q = 2.10 → q = 14 quarters
6	Guard A leaves the door walking 2 m per second. Guard B leaves the same door 9 seconds later, walking 3.5 m per second in the same direction. How…	2(t + 9) = 3.5t → 2t + 18 = 3.5t → 18 = 1.5t → t = 12 seconds
7	For what value of m does mx + 5 = 4x − 1 have no solution? Explain why.	m = 4: then 4x + 5 = 4x − 1 → 5 = −1, which is false, so there is no solution.
8	Prove: ax + b = cx + d has exactly one solution when a ≠ c. Find a formula for that solution.	ax − cx = d − b → (a − c)x = d − b. Since a ≠ c, a − c ≠ 0, so we can divide: x = (d − b) ÷ (a − c), one solution.

Part 3: Design Your Own Room

#	Task	Answer
9	Write a story and pick your codes first. Your five locks must include: a rational equation, a literal equation (solve a formula), a parameter…	Answers vary.
10	Write a full answer key with every step justified (name the property you use).	Answers vary.
11	Swap with another group. Record their time and the hardest lock. Revise one lock so that it is more challenging but still has a single integer…	Answers vary; integer code guaranteed when (a − c) divides (d − b).

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Locks are correctly chained and meet every equation requirement. Goes beyond the minimum with a creative, realistic choice.	Locks are correctly chained and meet every equation requirement.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Escape room kit includes a story, lock cards, an answer key, and test results. Polished, organized, and visually engaging.	Escape room kit includes a story, lock cards, an answer key, and test results. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.